

3 hrs

10 / Jun / 2025

Robert Sobukwe 230

EXAMS OFFICE
USE ONLY

University of the Witwatersrand, Johannesburg

Course or topic No(s)

COMS4062A/7074A

Course or topic name(s)
Paper Number & title

Probabilistic Graphical Models

Examination to be held during the month(s) of

June 2025

Year of study

1

Degrees/Diplomas for which this course is prescribed

BScHons (CS / BDA / CAM), MSc (AI / DS / CS / Robotics/ e-Science)

Faculties presenting candidates

Science

Internal examiner(s)

Prof. Ritesh Ajoodha
x-76188

External examiner(s)

Prof. JH Van Vuuren (Stellenbosch Uni.)

Special materials

Formula sheet and non-programmable calculator permitted

Time allowance

3 hours

Instructions to candidates

See next page.

Student Details

Please fill in the details below:

Student Number: _____

Student Name: _____

Exam Instructions: COMS7074A students are required to answer all questions. For COMS4062A, answer all questions except for Question 6. Answers written outside this booklet will not be graded. This is a closed book exam that spans 22 pages. There is a total of 90 marks available for COMS7074A, and 70 marks available for COMS4062A.

Mark Table:

Component	Grade
Question 1: MCQ Questions (10 Marks)	
Question 2: Inference and Learning (30 Marks)	
Question 3: Causality (10 Marks)	
Question 4: Structured Decision Problems (10 Marks)	
Question 5: General Problem Solving (10 Marks)	
Question 6: COMS7074A Additional Question (20 Marks)	

Question 1**MCQ Questions****[10 Marks]**

1. For each of the following MCQ questions, circle the correct answer label.
- 1.1. What does the Chernoff bound quantify in the context of sampling-based estimation? [2]
- (a) The expected log-likelihood of a Bayesian network.
 - (b) The rate at which the variance of a sample mean increases.
 - (c) The exponential decay in the probability that the sample mean deviates significantly from the expected value. (Answer)
 - (d) The upper bound on the number of particles required in Gibbs sampling.
- 1.2. In Bayesian inference, how does a sharply peaked prior distribution influence posterior outcomes? [2]
- (a) It guarantees convergence to the true model with fewer observations.
 - (b) It acts as a strong belief, significantly constraining the influence of new data. (Answer)
 - (c) It increases model flexibility, allowing for more varied posterior predictions.
 - (d) It always results in overfitting to the training data.
- 1.3. The Bayesian Information Criterion (BIC) penalises which of the following aspects during model selection? [2]
- (a) Data sparsity.
 - (b) Model complexity, such as the number of parameters. (Answer)
 - (c) The amount of noise in the data.
 - (d) The training error rate.
- 1.4. Why is the true causal graph $\mathcal{G}^*$ generally unidentifiable from the observed distribution P^* ? [2]
- (a) Because most distributions do not satisfy the Causal Markov Condition.
 - (b) Because multiple causal graphs can encode the same set of independencies. (Answer)
 - (c) Because intervention queries are always required for identification.
 - (d) Because observational data always contains selection bias.
- 1.5. In a structured decision problem, a complete strategy is best described as: [2]
- (a) A static rule that maximizes utility without considering state information.
 - (b) A utility-maximizing heuristic that only uses partial evidence.
 - (c) A mapping from all possible information states to decisions. (Answer)
 - (d) A stochastic function for selecting actions based on rewards.

Question 2**Inference and Learning****[30 Marks]**

- 2.1. [Figure 1](#) illustrates a dataset and a Bayesian network that encodes the probabilistic relationship between animal species (denoted A) and fur colour (denoted F). The dataset is drawn from a true underlying distribution P^* and categorises animal species into two types: rabbit (a^0) and cat (a^1). Fur colour is likewise divided into white (f^0) and black (f^1). Observations of fur colour were collected in a natural habitat and used to construct the variable F . Use the dataset and the Bayesian network provided in [Figure 1](#) to answer the following questions.

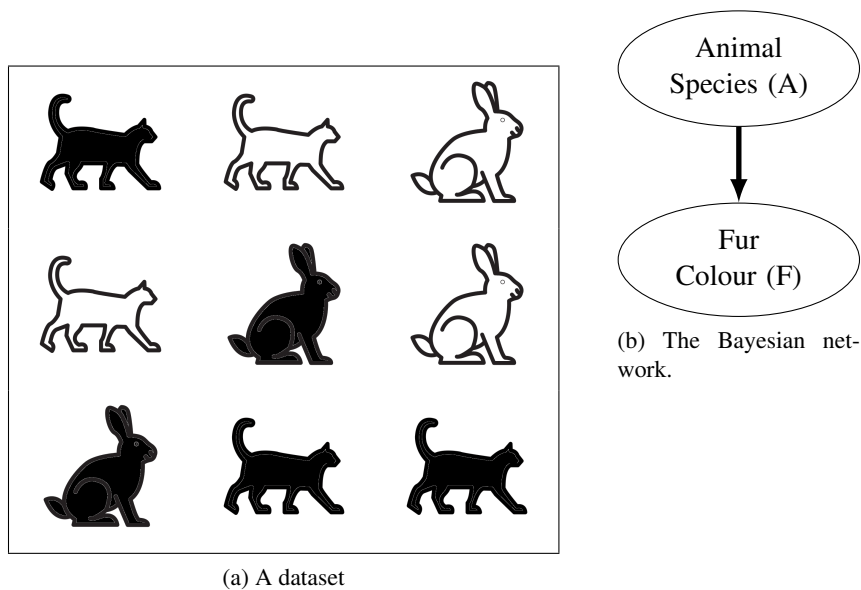


Figure 1: A data set (a) and Bayesian network (b)

- (a) Compute the maximum likelihood estimate: $P(a^0)$, $P(f^0 | a^0)$, $P(f^0 | a^1)$, and $P(f^1 | a^1)$. Show your working and round off your answer two decimal places. [4]

- (b) Using a Dirichlet prior, where $\alpha = 3$, compute the Bayesian estimate $P(a^0)$, $P(f^0 | a^0)$, $P(f^0 | a^1)$, and $P(f^1 | a^1)$. Show your working and round off your answer three decimal places. [4]

- (c) What does it mean for a model to generalise well in the context of probabilistic graphical models? Suggest a method to assess generalisation performance when P^* is unknown. [2]

- 2.2. Use the following Markov chain which describes the distribution between 3 states x^1 , x^2 , and x^3 , where x^3 is the initial state, to answer the following questions.

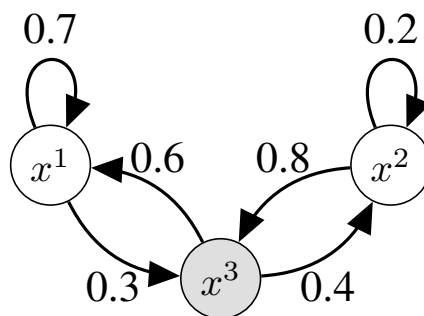


Figure 2: A Markov chain with 3 states.

- (a) Write the system of linear equations that governs the stationary distribution for the given system. [2]

-
- (b) Use Gauss-Jordan elimination (or any procedure of your choice) to calculate the stationary distribution: $\pi(x^1)$, $\pi(x^2)$, and $\pi(x^3)$. [6]

-
- (c) Demonstrate that your stationary distribution provided in the previous question satisfies detailed balance between x^1 and x^3 . [2]
-

- 2.3. Suppose you are given the following set of animals as depicted in Table 1. Each animal can either be a cat or a rabbit.

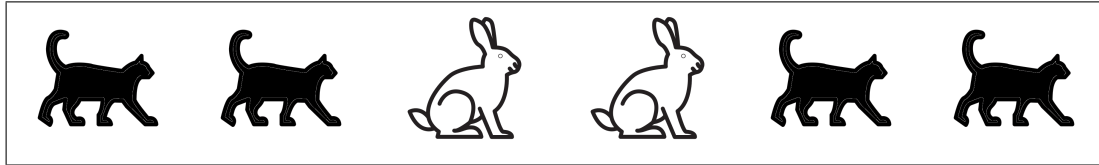


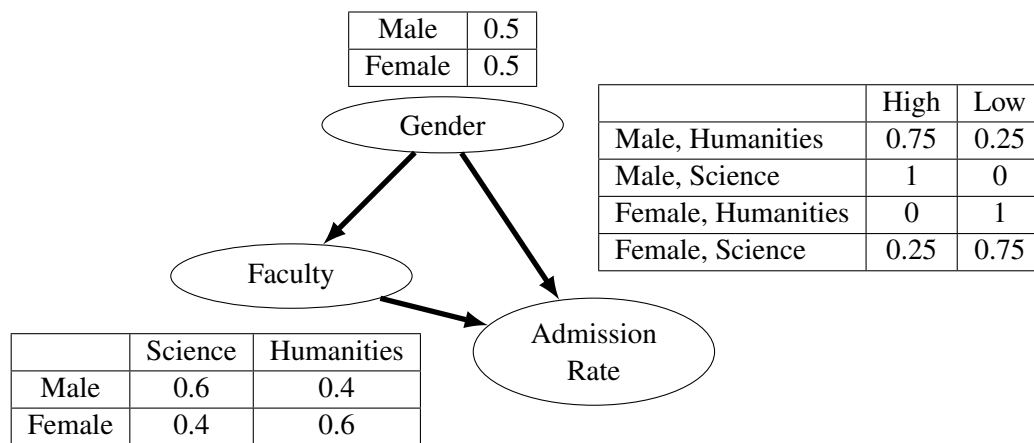
Table 1: A dataset

- (a) Calculate the marginal likelihood, $P(x[1], \dots, x[6])$, of the set of animals with a Dirichlet $[\alpha_0, \alpha_1]$ prior, where $\alpha_0 = 2$ represents cats, and $\alpha_1 = 3$ represents rabbits. Round off your answer two decimal places. [6]

- (b) Exact inference is known to be intractable in the worst case. One popular approach to circumvent this challenge is *variable elimination*. Briefly explain the idea behind variable elimination and why elimination ordering is important. [4]

Question 3**Causal Graphical Models****[10 Marks]**

3. Suppose that you encounter the following scenario of Simpson's paradox. In a university with two faculties, Science and Humanities, female applicants have higher admission rates than males within each faculty. However, because the larger Humanities faculty has a much lower admission rate and a higher proportion of male applicants, the aggregated data across both faculties misleadingly suggests that males have a higher overall admission rate, demonstrating Simpson's Paradox. The causal model demonstrating this is provided in Figure 3.

Figure 3: A causal model, $\mathcal{C}$

- (a) To accurately determine which faculty has the higher admission rate, it is necessary to remove the association between gender and faculty. Draw the mutilated causal network structure and update the parameters for $\mathcal{C}_F$, representing an intervention where the faculty variable is set independently of any other influencing causal factors. [3]

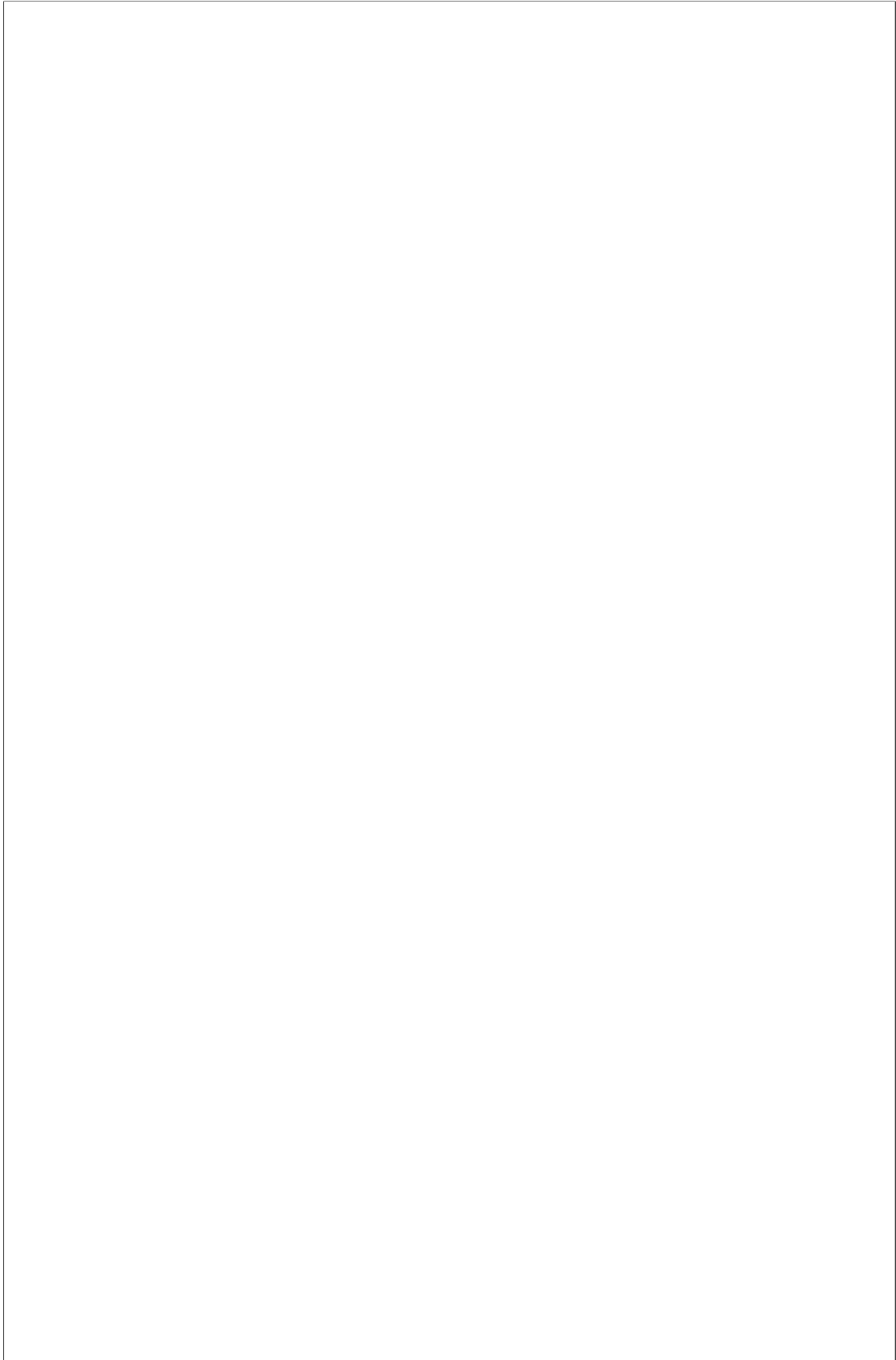
- (b) Given the mutilated network and updated parameters, you want to compare the probability of a high admission rate in both faculties (Science and Humanities). Write

the intervention query for this inference task and simplify it using the Iterated Query Simplification rules to eliminate the intervention in the query. [3]

- (c) Finally, calculate the probability of a high admission rate in both faculties using the updated parameters of the mutilated network. [4]

[illegible]

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.



END OF EXAM

Additional writing paper

[illegible]

[illegible]

Probabilistic Graphical Models

Formula Sheet

Probability Theory

Chain Rule for Probabilities:

$$P(X_1, \dots, X_n) = P(X_1) \dots P(X_n | X_1, X_{n-1})$$

Bayes Rule:

$$P(\alpha | \beta) = \frac{P(\beta | \alpha)}{P(\alpha)P(\beta)}$$

Probability Density Function: $p : \mathbb{R} \rightarrow \mathbb{R}$ is a

probability density function (PDF) for $\mathcal{X}$ if it is a non-negative integrable function such that:

$$\int_{\mathcal{V}al(\mathcal{X})} p(x) dx = 1.$$

Uniform Distribution: $X \sim \text{Unif}[a, b]$ if it has the PDF:

$$p(x) = \begin{cases} \frac{1}{b-a} & b \geq x \geq a \\ 0 & \text{otherwise.} \end{cases}$$

Gaussian Distribution: X has a Gaussian

distribution: $X \sim \mathcal{N}(\mu; \sigma^2)$ if it has the PDF:

$$p(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}.$$

Joint Density Function: Let P be a joint

distribution over $X_1, \dots, X_n$. A function

$p(x_1, \dots, x_n)$ is a joint density function of $X_1, \dots, X_n$ if:

1. $p(x_1, \dots, x_n) \geq 0 \forall x_1, \dots, x_n \in X_1, \dots, X_n$.
2. p is integratable.
3. For any choice of $a_1, \dots, a_n$ and $b_1, \dots, b_n$:

$$P(a_1 \leq X_1 \leq b_1, \dots, a_n \leq X_n \leq b_n) \\ = \int_{a_1}^{b_1} \dots \int_{a_n}^{b_n} p(x_1, \dots, x_n) dx_1 \dots dx_n$$

Conditional Density Function: Suppose you would like to condition over the event:

$$x - \epsilon \leq X \leq x + \epsilon.$$

Then

$$P(Y | x) = \lim_{\epsilon \rightarrow 0} P(Y | x - \epsilon \leq X \leq x + \epsilon).$$

If there is a continuous joint density function $p(x, y)$ then

$$P(a \leq Y \leq b | x - \epsilon \leq X \leq x + \epsilon) \\ = \frac{P(a \leq Y \leq b, x - \epsilon \leq X \leq x + \epsilon)}{P(x - \epsilon \leq X \leq x + \epsilon)} = \frac{\int_a^b \int_{x-\epsilon}^{x+\epsilon} p(x', y) dy dx'}{\int_{x-\epsilon}^{x+\epsilon} p(x') dx'}$$

Expectation of X under P is:

$$\mathbb{E}_P[X] = \sum_x x.P(x).$$

Expectation if $\mathbf{X}$ is Continuous:

$$\mathbb{E}_P[X] = \int x.p(x) dx.$$

Linearity of Expectation:

$$\mathbb{E}_P[X + Y] = \mathbb{E}_P[X] + \mathbb{E}_P[Y].$$

Conditional Expectation:

$$\mathbb{E}_P[X | \mathbf{y}] = \sum_x x.P(x | \mathbf{y}).$$

Variance of $\mathbf{X}$:

$$\mathbb{V}ar_P[X] = \mathbb{E}_P[(X - \mathbb{E}_P[X])^2].$$

Standard Deviation:

$$\sigma_X = \sqrt{\mathbb{V}ar_P[X]}.$$

Expectation and Variance of Gaussian

distribution $X \sim \mathcal{N}(\mu; \sigma^2)$, then $\mathbb{E}[X] = \mu$ and

$$\mathbb{V}ar[X] = \sigma^2.$$

Graph Theory

A **Graph** is a data structure $\mathcal{K} = (\mathcal{X}, \mathcal{E})$ consisting

of a set of nodes, denoted $\mathcal{X} = X_1, \dots, X_n$, and

edges, denoted $\mathcal{E}$.

Induced Subgraph: Let $\mathcal{K} = (\mathcal{X}, \mathcal{E})$, and $\mathbf{X} \in \mathcal{X}$,

then an induced subgraph, denoted $\mathcal{K}[\mathbf{X}]$ is a graph

$(\mathbf{X}, \mathcal{E}')$ where $\mathcal{E}'$ are all the edges $X \rightleftharpoons Y \in \mathcal{E}'$ such

that $X, Y \in \mathbf{X}$.

Complete Graph (Clique): A subgraph over $\mathbf{X}$ is

complete if every two nodes in $\mathbf{X}$ are connected by

some edge. The set $\mathbf{X}$ is called a clique. A clique $\mathbf{X}$

is maximal if for any superset of nodes $\mathbf{Y} \supset \mathbf{X}$, $\mathbf{Y}$ is

not a clique.

Upward Closure: A subset of nodes $\mathbf{X} \in \mathcal{X}$ is

upwardly closed in $\mathcal{K}$ if, for any $\mathbf{X} \in \mathcal{X}$, we have that

the Boundary $\mathbf{x} \subset \mathbf{X}$. We define upward closure of $\mathbf{X}$

to be the minimally upward closed subset $\mathbf{Y}$ that

contains $\mathbf{X}$.

Topological ordering: An ordering of the nodes

$X_1, \dots, X_n$ is a topological ordering if when we have

$$(X_i \rightarrow X_j) \in \mathcal{E}, \text{ then } i < j.$$

Chordal Graph: Let $X_1 - X_2 - \dots - X_k - X_1$ be a

loop in a graph. A chord in a loop is an edge

connecting X_i and X_j for two nonconsecutive nodes

X_i, X_j . An undirected graph $\mathcal{H}$ is said to be chordal

if and loop $X_1 - X_2 - \dots - X_k - X_1$ for $k > 4$ has a

chord. A directed graph $\mathcal{K}$ is said to be chordal if its

underlying undirected graph is chordal.

Bayesian Networks

Naïve Bayes:

$$P(C, X_1, \dots, X_n) = P(C) \prod_{i=1}^n P(X_i | C)$$

Bayesian Network:

$$P(X_1, \dots, X_n) = \prod_{i=1}^n P(X_i | \text{Pa}_{X_i}^G)$$

Deterministic CPD: $f : \mathcal{V}al(P_{a_X}) \mapsto \mathcal{V}al(X)$ s.t.:

$$P(x | pa_x) = \begin{cases} 1 & \text{if } x = f(pa_x) \\ 0 & \text{if } x \text{ otherwise} \end{cases}$$

Time Granularity Assumption:

$$P(\mathcal{X}^{(0:T)}) = P(\mathcal{X}^{(0)}) \prod_{t=0}^{T-1} P(\mathcal{X}^{(t+1)} | \mathcal{X}^{(0:t)})$$

Markov Assumption:

$$P(\mathcal{X}^{(0:T)}) = P(\mathcal{X}^{(0)}) \prod_{t=0}^{T-1} P(\mathcal{X}^{(t+1)} | \mathcal{X}^{(t)})$$

Time Invariance Assumption:

$$P(\mathcal{X}^{(t+1)} = \xi' | \mathcal{X}^{(t)} = \xi) = P(\mathcal{X}' = \xi' | \mathcal{X} = \xi)$$

Two-TBN:

$$P(\mathcal{X}' | \mathcal{X}) = P(\mathcal{X}' | \mathcal{X}_t) = \prod_{i=1}^n P(X'_i | P_{a_{X'_i}})$$

Linear Dynamical Systems:

$$P(\mathbf{X}^{(t)} | \mathbf{X}^{(t-1)}) = \mathcal{N}(\mathbf{A}\mathbf{X}^{(t-1)}; Q)$$

$$P(O^{(t)} | \mathbf{X}^{(t)}) = \mathcal{N}(\mathbf{H}\mathbf{X}^{(t)}; R)$$

Gibbs Distribution: A distribution P_Φ is a Gibbs

distribution parameterised by a set of factors

$\Phi = \{\phi_1(\mathbf{D}_1), \dots, \phi_K(\mathbf{D}_K)\}$ if it is defined as:

$$P_\Phi(X_1, \dots, X_n) = \frac{1}{Z} P_\Phi(X_1, \dots, X_n)$$

Inference

Inference:

$$P(\mathbf{Y} | \mathbf{E} = \mathbf{e}) = \frac{P(\mathbf{Y}, \mathbf{e})}{P(\mathbf{e})} = \frac{\sum_w P(\mathbf{y}, \mathbf{e}, \mathbf{w})}{\sum_{y', w} P(\mathbf{e})}$$

Sum-Product Message Passing:

$$\delta_{i \rightarrow j} = \sum \mathbf{C}_i - \mathbf{s}_{i,j} (\psi_i \times \prod_{k \in (N\mathbf{b}_i - \{j\})} \delta_{k \rightarrow i})$$

Tree Calibration:

$$\sum \mathbf{C}_i - \mathbf{s}_{i,j} \beta_i(\mathbf{C}_i) = \sum \mathbf{C}_j - \mathbf{s}_{i,j} \beta_j(\mathbf{C}_j)$$

Graph Calibration:

$$\sum \mathbf{C}_i - \mathbf{s}_{i,j} \beta_i = \sum \mathbf{C}_j - \mathbf{s}_{i,j} \beta_j$$

MAP:

$$\text{MAP}(\mathbf{Y} = \mathbf{y} | \mathbf{E} = \mathbf{e})$$

$$= \text{argmax}_{\mathbf{y}} P(\mathbf{Y} = \mathbf{y} | \mathbf{E} = \mathbf{e})$$

Convergence Bound:

$$\mathbb{E}_{\mathcal{D}}(f) = \frac{1}{M} \sum_{m=1}^M f(\xi[m]).$$

Hoeffding Bound:

$$P_{\mathcal{D}}(\hat{P}(\mathbf{y}) \notin [P(\mathbf{y}) - \epsilon, P(\mathbf{y}) + \epsilon]) \leq 2e^{-2M\epsilon^2}$$

Chernoff Bound:

$$P_{\mathcal{D}}(\hat{P}(\mathbf{y}) \notin [P(\mathbf{y})(\pm\epsilon)]) \leq 2e^{-MP(\mathbf{y})\epsilon^2/3}$$

$$M \geq 3 \frac{\ln(2/\delta)}{P(\mathbf{y})\epsilon^2}.$$

Likelihood Weighting:

$$\hat{P}_{\mathcal{D}}(\mathbf{y} \mid \mathbf{e}) = \frac{\sum_{m=1}^M w[m] \mathbb{1}\{\mathbf{y}[m]=\mathbf{y}\}}{\sum_{m=1}^M w[m]}.$$

MCMC Sampling:

$$P^{(t+1)}(\mathbf{X}^{(t+1)} = \mathbf{x}') =$$

$$\sum_{\mathbf{x} \in \text{Val}(\mathbf{x})} P^{(t)}(\mathbf{X}^{(t)} = \mathbf{x}) \mathcal{T}(\mathbf{x} \rightarrow \mathbf{x}')$$

Stationary Distribution:

$$\pi(\mathbf{X} = \mathbf{x}') = \sum_{\mathbf{x} \in \text{Val}(\mathbf{x})} \pi(\mathbf{X} = \mathbf{x}) \mathcal{T}(\mathbf{x} \rightarrow \mathbf{x}')$$

Detailed Balance Equation:

$$\pi(x) \mathcal{T}(x \rightarrow x') = \pi(x') \mathcal{T}(x' \rightarrow x)$$

Acceptance Probability:

$$\mathcal{A}(x \rightarrow x') = \min[1, \frac{\pi(x') \mathcal{T}^Q(x' \rightarrow x)}{\pi(x) \mathcal{T}^Q(x \rightarrow x')}].$$

Metropolis-Hastings Acceptance Probability:

$$\mathcal{A}(x_{-i}, x_i \rightarrow x_{-i}, x'_i) = \min[1, \frac{P_{\theta}(x'_i, x_{-i}) \mathcal{T}_{\theta}^Q(x_{-i}, x'_i \rightarrow x_{-i}, x_i)}{P_{\theta}(x_i, x_{-i}) \mathcal{T}_{\theta}^Q(x_{-i}, x_i \rightarrow x_{-i}, x'_i)}].$$

Learning

Relative Entropy:

$$\mathbb{D}(P^* \parallel \tilde{P}) = \mathbb{E}_{\xi \sim P^*} [\log(\frac{P^*(\xi)}{\tilde{P}(\xi)}),$$

Negative Empirical Log-loss:

$$\log P(\mathcal{D} : \mathcal{M}) = \sum_{m=1}^M \log P(\xi[m] : \mathcal{M}).$$

Bayesian Parameter Estimation:

$$P(\theta \mid x[1], \dots, x[M]) = \frac{P(x[1], \dots, x[M] \mid \theta) P(\theta)}{P(x[1], \dots, x[M])}$$

Expected Sufficient Statistics:

$$\bar{M}_{\theta}[\mathbf{y}] = \sum_{m=1}^M \sum_{\mathbf{h}[m] \in \text{Val}(\mathbf{H}[m])} Q(\mathbf{h}[m]) \mathbb{1}\{\xi[m]\langle \mathbf{Y} \rangle = \mathbf{y}\}$$

Maximisation of Expected Parameter:

$$\tilde{\theta}_{d^1|c^0} = \frac{M_{\theta}(d^1, c^0)}{M_{\theta}(c^0)}$$

Bayesian Clustering:

$$\bar{M}_{\theta}[c] = \frac{\bar{M}_{\theta}[c]}{M}$$

$$\bar{M}_{\theta}[x_i \mid c] = \frac{\bar{M}_{\theta}[x_i, c]}{\bar{M}_{\theta}[c]}$$

K-means Clustering:

$$c[m] = \text{argmax}_c P(c \mid x[m], \theta^t)$$

Hypothesis Testing:

$$d_{\mathbb{H}}(\mathcal{D}) = \sum_{x,y} \frac{M[x,y]}{M} \log \frac{M[x,y]/M}{M[x]/M \cdot M[y]/M}$$

$$R_{d,t}(\mathcal{D}) \begin{cases} \text{Accept if } d(\mathcal{D}) \leq t \\ \text{Reject if } d(\mathcal{D}) > t \end{cases}$$

$$\text{p-value}(t) = P(\{\mathcal{D} : d(\mathcal{D}) > t\} \mid H_0, M)$$

Likelihood:

$$\begin{aligned} \mathbb{I}_{\hat{P}_{\mathcal{D}}}(\mathbf{X}_i; Pa_{\mathbf{X}_i}^{\mathcal{G}}) \\ = \sum_{\mathbf{u}_i} \sum_{\mathbf{x}_i} \hat{P}(x_i, \mathbf{u}_i) \log \frac{\hat{P}(x_i, \mathbf{u}_i)}{\hat{P}(x_i) \hat{P}(\mathbf{u}_i)} \end{aligned}$$

Entropy:

$$\mathbb{H}_{\hat{P}_{\mathcal{D}}}(\mathbf{X}_i) = \sum_{x_i} \hat{P}(x_i) \log \frac{1}{\hat{P}(x_i)}$$

Bayesian Structure Learning:

$$P(\mathcal{G} \mid \mathcal{D}) = \frac{P(\mathcal{D} \mid \mathcal{G}) P(\mathcal{G})}{P(\mathcal{D})}$$

$$\text{score}_{\mathcal{B}}(\mathcal{G} : \mathcal{D}) = \log P(\mathcal{D} \mid \mathcal{G}) + \log P(\mathcal{G})$$

$$P(\mathcal{D} \mid \mathcal{G}) = \int_{\Theta_{\mathcal{G}}} P(\mathcal{D} \mid \theta_{\mathcal{G}}, \mathcal{G}) P(\theta_{\mathcal{G}} \mid \mathcal{G}) d\theta_{\mathcal{G}}$$

Marginal Likelihood for Binomials:

$$P(x[1], \dots, x[M])$$

$$= P(x[1]) \cdot \dots \cdot P(x[M] \mid x[1], \dots, x[M-1])$$

Marginal Likelihood for Multinomials:

$$P(x[1], \dots, x[M]) = \frac{\Gamma(\alpha)}{\Gamma(\alpha+M)} \cdot \prod_{i=1}^k \frac{\Gamma(\alpha_i + M[x^i])}{\Gamma(\alpha_i)}$$

Bayesian Score:

$$P(\mathcal{D} \mid \mathcal{G}) = \prod_i \prod_{\mathbf{u}_i \in \text{Val}(Pa_{\mathbf{X}_i}^{\mathcal{G}})} \frac{\Gamma(\alpha_{\mathbf{X}_i|\mathbf{u}_i}^{\mathcal{G}})}{\Gamma(\alpha_{\mathbf{X}_i|\mathbf{u}_i}^{\mathcal{G}} + M[\mathbf{u}_i])}.$$

$$\prod_{\mathbf{x}_i^j \in \text{Val}(\mathbf{X}_i)} \left[\frac{\Gamma(\alpha_{\mathbf{X}_i|\mathbf{u}_i}^{\mathcal{G}} + M[x_i^j, \mathbf{u}_i])}{\Gamma(\alpha_{\mathbf{X}_i|\mathbf{u}_i}^{\mathcal{G}})} \right]$$

BIC Score:

$$\text{score}_{\text{BIC}}(\mathcal{G} : \mathcal{D}) =$$

$$M \sum_{i=1}^n \mathbb{I}_{\hat{P}_{\mathcal{D}}}(\mathbf{X}_i; Pa_{\mathbf{X}_i}^{\mathcal{G}}) - \frac{\log M}{2} \dim[\mathcal{G}]$$

Decomposability:

$$\text{score}(\mathcal{G} : \mathcal{D}) = \sum_i \text{FamScore}(\mathbf{X}_i \mid Pa_{\mathbf{X}_i}^{\mathcal{G}} : \mathcal{D})$$

Tree weight:

$$w_{i \rightarrow j} = \text{FamScore}(\mathbf{X}_i \mid \mathbf{X}_j : \mathcal{D}) -$$

$$\text{FamScore}(\mathbf{X}_i : \mathcal{D})$$

Learning Graphs:

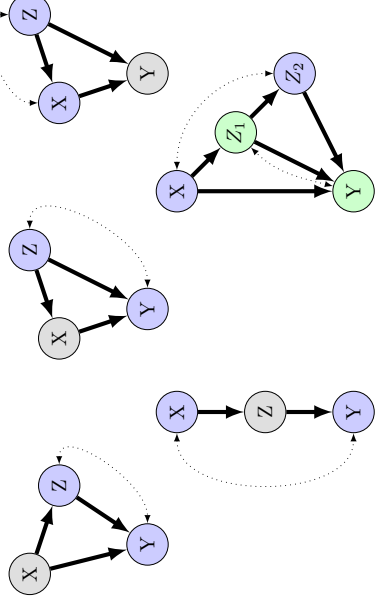
$$\mathcal{G}^* = \text{argmax}_{\mathcal{G} \in \mathcal{G}} \text{score}(\mathcal{G} : \mathcal{D})$$

Causality

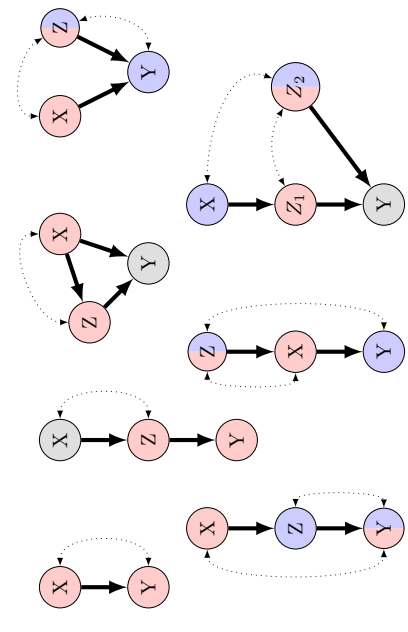
Intervention Query:

$$P_{\mathcal{C}}(\mathbf{Y} \mid do(z), \mathbf{x}) = P_{\mathcal{C}_{z=z}}(\mathbf{Y} \mid \mathbf{x})$$

Identifiable when $P(Y \mid do(X))$:



Not Identifiable when $P(Y \mid do(X))$:



Learning with Intervention Data:

$$P(\xi \mid do(\mathbf{Z} := \mathbf{z}), \mathcal{C}) = \prod_{X_i \notin \mathbf{Z}} P(x_i \mid \mathbf{u}_i)$$

Sufficient Statistics (Intervention Data):

$$M[x_i; \mathbf{u}_i] =$$

$$\sum_{m: X_i \notin \mathbf{Z}[m]} \mathbb{1}\{X_i[m] = x_i, Pa_{X_i}[m] = \mathbf{u}_i\}$$

Likelihood of Data (Intervention):

$$L(\mathcal{C} : \mathcal{D}) = \prod_{i=1}^n \prod_{x_i \in \text{Val}(X_i), \mathbf{u}_i \in \text{Val}(Pa_{X_i})} \theta_{x_i|\mathbf{u}_i}^{M[x_i; \mathbf{u}_i]}$$

Decision Theory

Expected Utility:

$$\text{EU}[D[a]] = \sum_{\mathbf{x}} P(\mathbf{x} \mid a) U(\mathbf{x}, a)$$

Maximum Expected Utility:

$$a^* = \text{argmax}_a \text{EU}[D[a]]$$

$$= \text{argmax}_a \sum_{\mathbf{x}} P(\mathbf{x} \mid a) U(\mathbf{x}, a)$$

Expected Utility with Information:

$$\text{EU}[D[\delta_A]] = \sum_{\mathbf{x}, a} P_{\delta_A}(\mathbf{x}, a) U(\mathbf{x}, a)$$

Maximal Expected Utility (MEU) Strategy:

$$\text{argmax}_{\delta_{D_1}, \dots, \delta_{D_k}} \text{EU}[\mathcal{Z}[\delta_{D_1}, \dots, \delta_{D_k}]]$$

Value of Information:

$$\text{VPI}(A \mid X) := \text{MEU}(D_{X \rightarrow A}) - \text{MEU}(D)$$